

Advanced Linear Algebra: Foundations to Frontiers

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1.2.6 Equivalence of vector norms ¶

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Homework 1.2.6.1. Fill out the following table:

x	$\ x\ _1$	$\ x\ _\infty$	$\ x\ _2$
$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$			
$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$			
$\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$			

► **Solution**

In this course, norms are going to be used to reason that vectors are "small" or "large". It would be unfortunate if a vector were small in one norm yet large in another norm. Fortunately, the following theorem excludes this possibility:

Theorem 1.2.6.1. Equivalence of vector norms. *Let $\|\cdot\| : \mathbb{C}^m \rightarrow \mathbb{R}$ and $|||\cdot||| : \mathbb{C}^m \rightarrow \mathbb{R}$ both be vector norms. Then there exist positive scalars σ and τ such that for all $x \in \mathbb{C}^m$*

$$\sigma \|x\| \leq |||x||| \leq \tau \|x\|.$$

Proof.

The proof depends on a result from real analysis (sometimes called "advanced calculus") that states that $\sup_{x \in S} f(x)$ is attained for some vector $x \in S$ as long as f is continuous and S is a compact (closed and bounded) set. For any norm $\|\cdot\|$, the unit ball $\|x\| = 1$ is a compact set. When a supremum is an element in S , it is called the maximum instead and $\sup_{x \in S} f(x)$ can be restated as $\max_{x \in S} f(x)$.

Those who have not studied real analysis (which is not a prerequisite for this course) have to take this on faith. It is a result that we will use a few times in our discussion.

We prove that there exists a τ such that for all $x \in \mathbb{C}^m$

$$|||x||| \leq \tau \|x\|,$$

leaving the rest of the proof as an exercise.

Let $x \in \mathbb{C}^m$ be an arbitrary vector. W.l.o.g. assume that $x \neq 0$. Then

$$\begin{aligned} & |||x||| \\ &= \quad < \text{algebra} > \\ & \frac{|||x|||}{\|x\|} \|x\| \\ &\leq \quad < \text{algebra} > \\ & \left(\sup_{z \neq 0} \frac{|||z|||}{\|z\|} \right) \|x\| \\ &= \quad < \text{change of variables: } y = z/\|z\| > \\ & \left(\sup_{\|y\|=1} |||y||| \right) \|x\| \\ &= \quad < \text{the set } \|y\| = 1 \text{ is compact} > \\ & \left(\max_{\|y\|=1} |||y||| \right) \|x\| \end{aligned}$$

The desired τ can now be chosen to equal $\max_{\|y\|=1} |||y|||$.

Homework 1.2.6.2. Complete the proof of [Theorem 1.2.6.1](#).

▼ [Solution](#)

We need to prove that

$$\sigma \|x\| \leq |||x|||.$$

From the first part of the proof of [Theorem 1.2.6.1](#), we know that there exists a $\rho > 0$ such that

$$\|x\| \leq \rho |||x|||$$

and hence

$$\frac{1}{\rho} \|x\| \leq |||x|||.$$

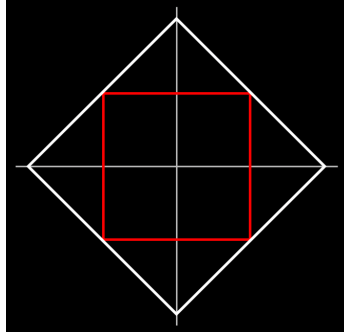
We conclude that

$$\sigma \|x\| \leq |||x|||$$

where $\sigma = 1/\rho$.

Example 1.2.6.2.

- Let $x \in \mathbb{R}^2$. Use the picture

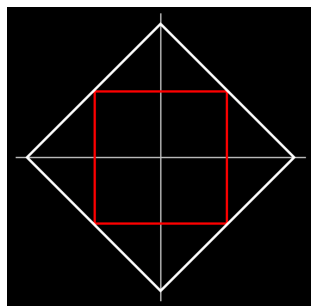


to determine the constant C such that $\|x\|_1 \leq C\|x\|_\infty$. Give a vector x for which $\|x\|_1 = C\|x\|_\infty$.

- For $x \in \mathbb{R}^2$ and the C you determined in the first part of this problem, prove that $\|x\|_1 \leq C\|x\|_\infty$.
- Let $x \in \mathbb{C}^m$. Extrapolate from the last part the constant C such that $\|x\|_1 \leq C\|x\|_\infty$ and then prove the inequality. Give a vector x for which $\|x\|_1 = C\|x\|_\infty$.

▼ Solution

- Consider the picture



- The red square represents all vectors such that $\|x\|_\infty = 1$ and the white square represents all vectors such that $\|x\|_1 = 2$.
- All points on or outside the red square represent vectors y such that $\|y\|_\infty \geq 1$. Hence *if $\|y\|_1 = 2$ then $\|y\|_\infty \geq 1$* .
- Now, pick any $z \neq 0$. Then $\|2z/\|z\|_1\|_1 = 2$. Hence

$$\|2z/\|z\|_1\|_\infty \geq 1$$

which can be rewritten as

$$\|z\|_1 \leq 2\|z\|_\infty.$$

Thus, $C = 2$ works.

- Now, from the picture it is clear that $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ has the property that $\|x\|_1 = 2\|x\|_\infty$. Thus, the inequality is "tight."

- We now prove that $\|x\|_1 \leq 2\|x\|_\infty$ for $x \in \mathbb{R}^2$:

$$\begin{aligned}
& \|x\|_1 \\
&= \quad < \text{definition} > \\
& |\chi_0| + |\chi_1| \\
&\leq \quad < \text{algebra} > \\
& \max(|\chi_0|, |\chi_1|) + \max(|\chi_0|, |\chi_1|) \\
&= \quad < \text{algebra} > \\
& 2 \max(|\chi_0|, |\chi_1|) \\
&= \quad < \text{definition} > \\
& 2\|x\|_\infty.
\end{aligned}$$

- From the last part we extrapolate that $\|x\|_1 \leq m\|x\|_\infty$.

$$\begin{aligned}
& \|x\|_1 \\
&= \quad < \text{definition} > \\
& \sum_{i=0}^{m-1} |\chi_i| \\
&\leq \quad < \text{algebra} > \\
& \sum_{i=0}^{m-1} \left(\max_{j=0}^{m-1} |\chi_j| \right) \\
&= \quad < \text{algebra} > \\
& m \max_{j=0}^{m-1} |\chi_j| \\
&= \quad < \text{definition} > \\
& m\|x\|_\infty.
\end{aligned}$$

Equality holds (i.e., $\|x\|_1 = m\|x\|_\infty$) for $x = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$.

Some will be able to go straight for the general result, while others will want to seek inspiration from the picture and/or the specialized case where $x \in \mathbb{R}^2$.

Homework 1.2.6.3. Let $x \in \mathbb{C}^m$. The following table organizes the various bounds:

	$\ x\ _1 \leq C_{1,2}\ x\ _2$	$\ x\ _1 \leq C_{1,\infty}\ x\ _\infty$
$\ x\ _2 \leq C_{2,1}\ x\ _1$		$\ x\ _2 \leq C_{2,\infty}\ x\ _\infty$
$\ x\ _\infty \leq C_{\infty,1}\ x\ _1$	$\ x\ _\infty \leq C_{\infty,2}\ x\ _2$	

For each, determine the constant $C_{x,y}$ and prove the inequality, including that it is a tight inequality.

Hint: look at the hint!

- ▶ [Hint](#)
- ▼ [Solution 1 \$\|x\|_1 \leq C_{1,2}\|x\|_2\$](#)
- ▼ [Solution 2 \$\|x\|_1 \leq C_{1,\infty}\|x\|_\infty\$](#)
- ▼ [Solution 3 \$\|x\|_2 \leq C_{2,1}\|x\|_1\$](#)
- ▼ [Solution 4 \$\|x\|_2 \leq C_{2,\infty}\|x\|_\infty\$](#)
- ▼ [Solution 5 \$\|x\|_\infty \leq C_{\infty,1}\|x\|_1\$](#)
- ▼ [Solution 6 \$\|x\|_\infty \leq C_{\infty,2}\|x\|_2\$](#)
- ▼ [Solution 7 Table of constants](#)

	$\ x\ _1 \leq \sqrt{m}\ x\ _2$	$\ x\ _1 \leq m\ x\ _\infty$
$\ x\ _2 \leq \ x\ _1$		$\ x\ _2 \leq \sqrt{m}\ x\ _\infty$
$\ x\ _\infty \leq \ x\ _1$	$\ x\ _\infty \leq \ x\ _2$	

$\|x\|_\infty \leq \|x\|_2$:

$$\begin{aligned}
 & \|x\|_\infty^2 \\
 &= \text{< definition >} \\
 & \left(\max_{i=0}^{m-1} |\chi_i| \right)^2 \\
 &= \text{< algebra >} \\
 & \max_{i=0}^{m-1} |\chi_i|^2 \\
 & \leq \text{< algebra >} \\
 & \sum_{i=0}^{m-1} |\chi_i|^2 \\
 &= \text{< definition >} \\
 & \|x\|_2^2.
 \end{aligned}$$

Taking the square root of both sides yields $\|x\|_\infty \leq \|x\|_2$.

Consider

$$x = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Then $\|x\|_\infty = 1 = \|x\|_2$.

$\|x\|_\infty \leq \|x\|_1$:

$$\begin{aligned}
 & \|x\|_\infty \\
 &= \text{< definition >} \\
 & \max_{i=0}^{m-1} |\chi_i| \\
 & \leq \text{< algebra >} \\
 & \sum_{i=0}^{m-1} |\chi_i| \\
 &= \text{< definition >} \\
 & \|x\|_1.
 \end{aligned}$$

Consider

$$x = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Then $\|x\|_\infty = 1 = \|x\|_1$.

$$\|x\|_2 \leq \sqrt{m}\|x\|_\infty:$$

$$\begin{aligned} \|x\|_2^2 &= \text{< definition >} \\ \sum_{i=0}^{m-1} |\chi_i|^2 &\leq \text{< algebra >} \\ \sum_{i=0}^{m-1} \left(\max_{j=0}^{m-1} |\chi_j| \right)^2 &= \text{< definition >} \\ \sum_{i=0}^{m-1} \|x\|_\infty^2 &= \text{< algebra >} \\ m\|x\|_\infty^2. \end{aligned}$$

Taking the square root of both sides yields $\|x\|_2 \leq \sqrt{m}\|x\|_\infty$.

Consider

$$x = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

then $\|x\|_2 = \sqrt{m}$ and $\|x\|_\infty = 1$ so that $\|x\|_2 = \sqrt{m}\|x\|_\infty$.

$$\|x\|_2 \leq \|x\|_1:$$

$$\begin{aligned} \|x\|_2^2 &= \text{< definition >} \\ \sum_{i=0}^{m-1} |\chi_i|^2 &\leq \text{< algebra >} \\ \left(\sum_{i=0}^{m-1} |\chi_i| \right)^2 &= \text{< definition >} \\ \|x\|_1^2. \end{aligned}$$

Taking the square root of both sides yields $\|x\|_2 \leq \|x\|_1$.

If we now choose

$$x = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

then $\|x\|_2 = \|x\|_1$.

$$\|x\|_1 \leq m\|x\|_\infty:$$

See [Example 1.2.6.2](#).

$$\|x\|_1 \leq \sqrt{m}\|x\|_2:$$

Consider $y = \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{m-1} \end{pmatrix} = \begin{pmatrix} \chi_0/|\chi_0| \\ \vdots \\ \chi_{m-1}/|\chi_{m-1}| \end{pmatrix}$. Then

$$|x^H y| = \left| \sum_{i=0}^{m-1} \overline{\chi_i} \chi_i / |\chi_i| \right| = \left| \sum_{i=0}^{m-1} |\chi_i|^2 / |\chi_i| \right| = \left| \sum_{i=0}^{m-1} |\chi_i| \right| = \|x\|_1.$$

We also notice that $\|y\|_2 = \sqrt{m}$.

From the Cauchy-Swartz inequality we know that

$$\|x\|_1 = |x^H y| \leq \|x\|_2 \|y\|_2 = \sqrt{m} \|x\|_2.$$

The problem with the above argument is that one of more χ_i may equal zero. The argument can be fixed by choosing

$$\psi_i = \begin{cases} \chi_i/|\chi_i| & \text{if } \chi_i \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

or

$$\psi_i = \begin{cases} \chi_i/|\chi_i| & \text{if } \chi_i \neq 0 \\ 1 & \text{otherwise.} \end{cases}$$

To demonstrate that equality is attained for at least one non-zero vector, we choose

$$x = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

then $\|x\|_1 = m$ and $\|x\|_2 = \sqrt{m}$ so that $\|x\|_1 = \sqrt{m}\|x\|_2$.

 **Remark 1.2.6.3.** The bottom line is that, modulo a constant factor, if a vector is "small" in one norm, it is "small" in all other norms. If it is "large" in one norm, it is "large" in all other norms.